

SUMMER RESEARCH INTERNSHIP PROGRAMME 2026

FINAL RESEARCH REPORT — SUBMISSION COVER PAGE

PROJECT / RESEARCH TITLE

Crop Yield Prediction for West Bengal for 2026–2027 Using Random Forest Regression

PROGRAMME DETAILS

Domain / Track: AI & Machine Learning Duration: 15/05/2026-15/08/2026

Department: ECE Batch / Year: 2026

SUBMITTED BY

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UNDER THE MENTORSHIP OF

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DECLARATION

We hereby declare that this report is a record of genuine work carried out by us under the guidance of the above-mentioned mentor, as part of the IEEE SMC Student Branch Chapter, KGEC — Research Internship Programme 2026, and has not been submitted elsewhere for any other purpose.

Mentor's Signature

Programme Coordinator

IEEE SMC Student Branch Chapter, KGEC | Research Internship Programme 2026 | Date of Submission: [19 / 08 / 2026]

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CROP YIELD PREDICTION FOR WEST BENGAL FOR 2026–2027 USING RANDOM FOREST REGRESSION

Abstract

Crop yield prediction is one of the interesting applications of machine learning because having knowledge about the future yields can help take necessary steps to plan and manage the resources. This research study focuses on using Random Forest Regression to predict the crop yields for the years 2026 and 2027 in West Bengal based on the past data. The data is pre-processed by removing irrelevant information and keeping only the records relevant to West Bengal. In the set of past data for each combination of Crop Name and Season, years are divided into groups according to their value. Based on these groups, multiple time-based features are engineered from the Yield, including Year, yield lag 1 year, yield lag 2 years, yield mean for the previous 3 years, and yield percentage change from the previous years.

For each combination of Crop Name and Season that has sufficient data, the Random Forest Regression model is trained using the set of engineered features above. The random forest consists of 300 decision trees, and random_state is set to 42. The models are evaluated using the mean absolute error and R-squared score. Finally, the optimal model is chosen using all the past data, and future yields are predicted for 2026 and 2027 with this model. The same approach is applied to the insufficient data groups. According to the research results, it is possible to make different observations about crop yield prediction for the region depending on the combination of Crop Name and Season. For example, in the case of Oilseeds Kharif, an R squared value as high as 0.852 was achieved, while for several other combinations, the value was close to zero or even negative. Thus, it can be concluded that the sequence of past crop yields can be used to predict future ones and hence might be beneficial for future planning.

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1. INTRODUCTION:-

1.1 Background:-

Agriculture is an important component of food security as well as economic development. The productivity of the crops keeps on changing every year due to various reasons like production trends of past years, environmental factors, and agricultural practices. Yield estimation plays a very significant role in agricultural planning.

The yield estimation process has traditionally relied on historical data, field measurement, and expert knowledge. Given the large amount of data now available, machine learning algorithms provide an alternate avenue to explore for capturing historical patterns.

Since yield is a numeric value it is a regression problem. This paper uses historical data of agriculture yield of West Bengal to estimate the possible yield of 2026 and 2027 using Random Forest Regression.

1.2 Problem Statement:-

The primary challenge that is being dealt with in this project involves predicting future crop yields based on combinations of crops and seasons within the state of West Bengal.

This challenge can be mathematically stated as:

Historical Crop Yield Data + Feature Engineering → Random Forest Regression Algorithm → Future Crop Yield Predictions.

The target years in this project are **2026 and 2027**.

1.3 Aim of the Study:-

Objectives of the Project:

To frame a prediction model using Random Forest regression for predicting yields of the crops in the state of West Bengal for the years 2026 & 2027.

1.4 Scope of the Study:-

The experiment involves predicting crop yield in West Bengal based on the data on agriculture provided with the description of the problem.

Only yield-related features are used in the algorithm, whereas such factors as rainfall, temperature, soil, fertilizer, irrigation, and other contributors to crop yield are not taken into account.

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2. LITERATURE REVIEW:-

2.1 Crop Yield Prediction:-

Crop yield forecasting is basically predicting crop yield in future using past observations along with other related data. According to the type of data that is present, crop yield forecasting may be based on one or more of the following factors.

2.2 Machine Learning in Agriculture:-

Machine learning algorithms have the ability to detect complex patterns in agricultural data and have a wide range of agricultural applications such as crop recommendation, crop disease detection, agricultural monitoring, and yield prediction.

For yield prediction purposes, regression algorithms are appropriate since the yield is measured on a continuous scale.

2.3 Regression-Based Yield Prediction:-

Regression models try to establish a relationship between the predictors and the continuous response variable. Some popular models are linear regressions, decision tree regressions, random forest regressions, gradient boosting regressions, XGBoost, and artificial neural networks.

The selection of the model is based on certain data properties and the prediction purpose.

2.4 Random Forest Regression:-

Random Forest is the method of ensemble learning that uses a set of decision trees. Random Forest predicts in regression problems by averaging all predictions made by the trees.

Random Forest is capable of working with non-linear dependencies and interactions of the variables.

2.5 Existing Research:-

In this reference, Artificial Neural Network methodology is used to predict the crop yield in West Bengal. In this report, historical agriculture data consisting of crop names, season names, state names, and crop yield values are described, along with their preprocessing, modeling, evaluation, and prediction.

One-hot encoding, year transformation, feature scaling, ANN modeling, and crop yield prediction for the year 2026 is presented in this supplied report.

2.6 Research Gap:-

The key difference between the current study and the existing ANN technique is that the current study applies Random Forest Regression model, as well as develops historical time dependent features, including the following: yield lagging, moving average, and percentage change.

Besides, another distinguishing characteristic of the suggested methodology is that it applies Crop and Season-specific modeling as well as forecasts the future for 2026 and 2027 years.

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3. DESCRIPTION OF DATASET:-

3.1 Dataset Source:-

The above analysis makes use of the dataset named:

"State-wise Yield of Food Grains in Kharif and Rabi Seasons"

The aforementioned report clearly points out that the dataset comprises fields such as Year, State / Uts, Crop Name, Season, and Yield.(See Fig-1 & Fig-2)

The actual dataset filename used for the current implementation is:

"State-wise Yield of Food Grains in Kharif and Rabi seasons (4).csv"

3.2 Dataset Overview:-

The raw data comprises observations on farming from multiple states, crops, seasons, and different years (Fig-1).

In order to proceed with the present analysis, the raw dataset has been cleaned to select observations belonging to West Bengal alone. The code further drops observations with missing yield value and selects variables needed for modelling .

3.3 Dataset Attributes:-

Attribute	Explanation
Year	Year in which agricultural observation takes place
State/UTs	State or Union Territory
Name of Crop	Name of crop or crop groups
Season	Crop season/category
Yield	Crop yield in kg per ha

3.4 Target Variable — Crop Yield:-

The target variable used in the regression model is:

Yield (Kg/Ha)

It represents the crop yield corresponding to a particular crop, season, year, and state.

3.5 West Bengal Data Extraction:-

Filtering of the entire dataset has been carried out based on the State / Uts variable such that all observations from the state of West Bengal are selected.

These West Bengal observations become the starting point for our further modeling.

3.6 Crop and Season Distribution:-

The model works independently based on each Crop and Season combination that is available. There are 31 Crop and Season combinations in the given output of predictions.

These Crop and Season combinations comprise of crops like Bajra, Foodgrains, Gram, Groundnut, Linseed, Maize, Oilseeds, Pulses, Rapeseed & Mustard, Rice, Safflower, Soybean, Sunflower, Tur, and Wheat, among others.

Country	Year	State / Uts	Crop Name	Season	Additional Yield (UOM:Kg/Ha(Kilogramperhectare)), Scaling Factor:1
2	India	Agriculture All India	Bajra	Total	3rd Advani 1388
3	India	Agriculture All India	Foodgrains	Kharif	3rd Advani 2298
4	India	Agriculture All India	Foodgrains	Rabi	3rd Advani 2935
5	India	Agriculture All India	Foodgrains	Total	3rd Advani 2578
6	India	Agriculture All India	Gram	Total	3rd Advani 1160
7	India	Agriculture All India	Groundnut	Kharif	3rd Advani 2076
8	India	Agriculture All India	Groundnut	Rabi	3rd Advani 2005
9	India	Agriculture All India	Groundnut	Summer	3rd Advani 2019
10	India	Agriculture All India	Groundnut	Total	3rd Advani 2067
11	India	Agriculture All India	Linseed	Total	3rd Advani 633
12	India	Agriculture All India	Maize	Kharif	3rd Advani 2947
13	India	Agriculture All India	Maize	Rabi	3rd Advani 4999
14	India	Agriculture All India	Maize	Summer	3rd Advani 4425
15	India	Agriculture All India	Maize	Total	3rd Advani 3518
16	India	Agriculture All India	Oilseeds	Kharif	3rd Advani 1386
17	India	Agriculture All India	Oilseeds	Rabi	3rd Advani 1456
18	India	Agriculture All India	Oilseeds	Total	3rd Advani 1408
19	India	Agriculture All India	Pulses	Total	3rd Advani 914
20	India	Agriculture All India	Rapeseed	Total	3rd Advani 1461
21	India	Agriculture All India	Rice	Kharif	3rd Advani 2807
22	India	Agriculture All India	Rice	Rabi	3rd Advani 3503
23	India	Agriculture All India	Rice	Summer	3rd Advani 3265
24	India	Agriculture All India	Rice	Total	3rd Advani 2899
25	India	Agriculture All India	Safflower	Total	3rd Advani 756
26	India	Agriculture All India	Soybean	Total	3rd Advani 1172

Fig-1:- Dataset image 1

832	India	Agriculture West Beng	Bajra	Total	3rd Advani 407
833	India	Agriculture West Beng	Foodgrain	Kharif	3rd Advani 2713
834	India	Agriculture West Beng	Foodgrain	Rabi	3rd Advani 3401
835	India	Agriculture West Beng	Foodgrain	Total	3rd Advani 3064
836	India	Agriculture West Beng	Gram	Total	3rd Advani 1240
837	India	Agriculture West Beng	Groundnut	Kharif	3rd Advani 1560
838	India	Agriculture West Beng	Groundnut	Rabi	3rd Advani 1900
839	India	Agriculture West Beng	Groundnut	Summer	3rd Advani 3280
840	India	Agriculture West Beng	Groundnut	Total	3rd Advani 3046
841	India	Agriculture West Beng	Linseed	Total	3rd Advani 480
842	India	Agriculture West Beng	Maize	Kharif	3rd Advani 4547
843	India	Agriculture West Beng	Maize	Rabi	3rd Advani 7074
844	India	Agriculture West Beng	Maize	Summer	3rd Advani 7270
845	India	Agriculture West Beng	Maize	Total	3rd Advani 6940
846	India	Agriculture West Beng	Oilseeds	Kharif	3rd Advani 1410
847	India	Agriculture West Beng	Oilseeds	Rabi	3rd Advani 1251
848	India	Agriculture West Beng	Oilseeds	Total	3rd Advani 1350
849	India	Agriculture West Beng	Pulses	Total	3rd Advani 968
850	India	Agriculture West Beng	Rapeseed	Total	3rd Advani 1250
851	India	Agriculture West Beng	Rice	Kharif	3rd Advani 2736
852	India	Agriculture West Beng	Rice	Rabi	3rd Advani Estimates
853	India	Agriculture West Beng	Rice	Summer	3rd Advani 3617
854	India	Agriculture West Beng	Rice	Total	3rd Advani 2950
855	India	Agriculture West Beng	Safflower	Total	3rd Advani 350
856	India	Agriculture West Beng	Soybean	Total	3rd Advani 750
857	India	Agriculture West Beng	Sunflower	Kharif	3rd Advani Estimates

Fig-2:- Dataset image 2

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4. DATA PREPROCESSING AND FEATURE ENGINEERING:-

4.1 Data Loading:-

The agriculture dataset was loaded from the Pandas library.

The yield column was designated as the target variable and was renamed to Yield.

4.2 Extraction of Agricultural Year:-

The Year field originally contains text data. The regular expression is employed to find out the four-digit agricultural year and put it into a numeric field labeled as YearNum.

Irregular and blank years are deleted.

4.3 Sorting Historical Data:-

West Bengal's data are organized according to:

1. Crop name
2. Season
3. Year

Order is vital because of the presence of lag and rolling characteristics of the data set.

4.4 Feature Engineering:-

Five numerical features were engineered for every Crop and Season pair:

Feature	Description
year_idx	The sequential position of the observation in the crop-season record
lag1	The previous observation/year yield
lag2	The yield from two observations/years before
roll3_mean	The mean of the last three observations
pct_change	The relative yield change with respect to the previous observation

4.5 Feature Interpretation:-

year_idx: This variable captures the sequential occurrence of the data point in the time sequence of a Crop and Season combination.

lag1: This is the immediate preceding yield and tells us how well the crop has done recently.

lag2: This gives additional temporal information based on information collected from two past yields.

roll3_mean: This gives smoothed information regarding the past behaviour of yield based on the three most recent observations.

pct_change: This is the relative change in the past yield and gives information about whether there is an increase or decrease in the yield.

The above five features will give the model information about historical yield behaviour.

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5. PROPOSED APPROACH:-

5.1 Overall System Architecture:-

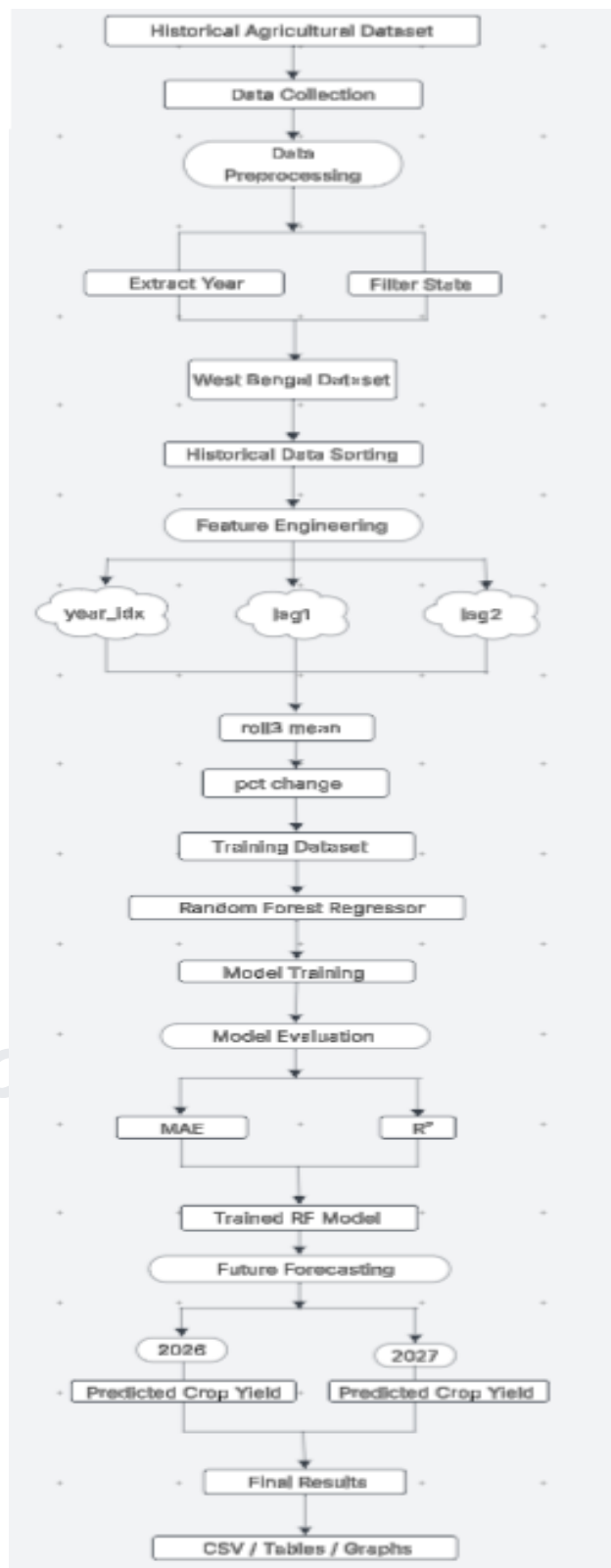


Fig 3:- Overall System Architecture

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5.2 Crop-Season Wise Modelling:-

Classification of the pre-processed data set for West Bengal takes place based on Crop Name and Season. For each well-sized crop category for season, a separate Random Forest model is created. In this manner, the model will have knowledge of the yield behaviour of each crop category.

5.4 Model Training:-

For groups that have enough historic data, the data set is split into a training and test set.

Test size – 25%

Random State - 42

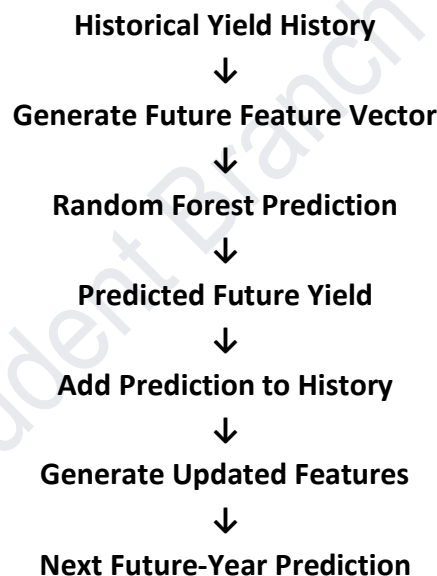
The Random Forest Evaluation model is trained on the training set.

The Final Random Forest model is later trained on all available historical data.

5.5 Recursive Forecasting Method:-

The study uses recursive forecasting for future prediction.

The forecasting logic can be represented as:



The recursive procedure allows the model to generate future values even when actual future yield values are unavailable.

5.6 Handling Insufficient Historical Data:-

This is done through providing a fall-back strategy for groups that lack enough history data during crop seasons. This is based on whether there are enough observations from history or not, thus allowing either the use of a linear trend or a persistence fall back instead of a faulty random forest. The method used by each group is documented in the results produced.

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6. RANDOM FOREST AND REGRESSION MODEL:-

6.1 Introduction to Random Forest Regression:-

Random Forest Regression is an ensemble learning algorithm that involves using different decision trees for training and making a combination of these predictions so as to make the final prediction. The use of different trees helps in reducing reliance on the particular decision tree.

6.2 Decision Trees:-

The decision tree algorithm uses rules for making decisions to create smaller spaces in the input space. In the case of regression, the terminal regions correspond to numbers that will be predicted.

6.3 Ensemble Learning:-

Fig-4 follows:

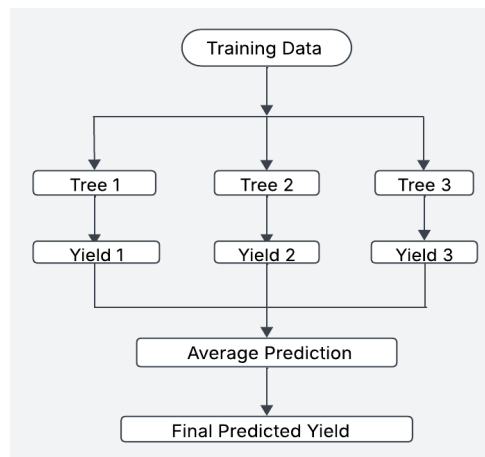


Fig 4:- Random Forest Regression architecture

6.4 Working Principle:-

This model employs several decision trees in order to understand how the generated features affect the output. These decisions are then combined to generate the final outcome for regression purposes.

6.5 Random Forest Architecture:-

For this project:

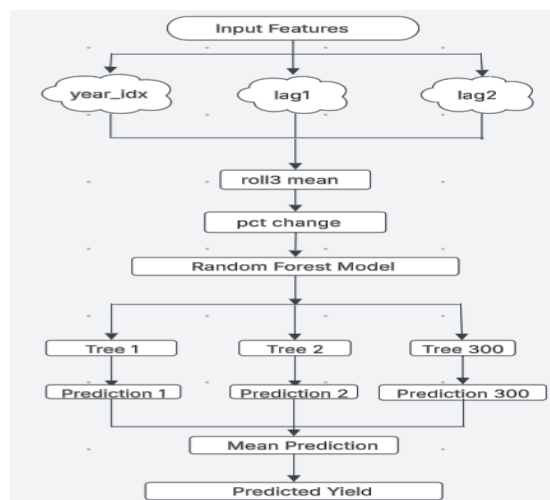


Fig 5:- Random Forest Architecture

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7. MODEL TRAINING AND EVALUATION:-

7.1 Code for predicted yield of 2026-2027:-

```
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score
import re
import warnings
warnings.filterwarnings("ignore")

# 1. Setup and Data Loading
INPUT_FILE = "State-wise Yield of Food Grains in Kharif and Rabi seasons (4).csv"
OUTPUT_FILE = "wb_yield_multiyear_recursive.csv"
df = pd.read_csv(INPUT_FILE)
YIELD_COL = "Yield (UOM:Kg/Ha(KilogramperHectare)), Scaling Factor:1"
df["YearNum"] = df["Year"].apply(lambda x: int(re.search(r"(\d{4})", str(x)).group(1)) if re.search(r"(\d{4})", str(x)) else np.nan)
df.dropna(subset=["YearNum"], inplace=True)
df["YearNum"] = df["YearNum"].astype(int)
wb = df[df["State / Uts"].str.contains("West Bengal", case=False, na=False)].copy()
wb = wb.dropna(subset=[YIELD_COL])
wb = wb[["Crop Name", "Season", "YearNum", YIELD_COL]].rename(columns={YIELD_COL: "Yield"})
def build_features(group):
    g = group.sort_values("YearNum").reset_index(drop=True)
    g["year_idx"] = np.arange(len(g))
    g["lag1"] = g["Yield"].shift(1)
    g["lag2"] = g["Yield"].shift(2)
    g["roll3_mean"] = g["Yield"].rolling(3, min_periods=1).mean().shift(1)
    g["pct_change"] = g["Yield"].pct_change().replace([np.inf, -np.inf], 0)
    return g
FEATURES = ["year_idx", "lag1", "lag2", "roll3_mean", "pct_change"]
results = []

# 2. Train + Forecast per group
for (crop, season), group in wb.groupby(["Crop Name", "Season"]):
    g = build_features(group)
    last_year = g["YearNum"].max()
    if len(g) < 6:
        gap_to_2027 = 2027 - last_year
        if len(g) >= 4 and gap_to_2027 <= 6:
            x = g["YearNum"].values
            y = g["Yield"].values
            coeffs = np.polyfit(x, y, 1)
            pred_2025 = np.polyval(coeffs, 2025)
            pred_2026 = np.polyval(coeffs, 2026)
            pred_2027 = np.polyval(coeffs, 2027)
            method = "Linear trend (insufficient data for RF)"
```

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```

else:
    pred_2025 = pred_2026 = pred_2027 = g["Yield"].iloc[-1]
    method = "Persistence Fallback"
results.append({
    "Crop Name": crop, "Season": season, "Last_Available_Year": 2025,
    "Last_Available_Yield (2025 Pred)": round(max(pred_2025, 0), 1),
    "Predicted_Yield_2026": round(max(pred_2026, 0), 1),
    "Predicted_Yield_2027": round(max(pred_2027, 0), 1),
    "Method": method, "Test_MAE": np.nan, "Test_R2": np.nan
})
continue
g_feat = g.dropna(subset=FEATURES)
X = g_feat[FEATURES]
y = g_feat["Yield"]

# 3. Split for metrics
if len(X) >= 8:
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=42)
    eval_model = RandomForestRegressor(n_estimators=300, random_state=42)
    eval_model.fit(X_train, y_train)
    test_preds = eval_model.predict(X_test)
    mae = mean_absolute_error(y_test, test_preds)
    r2 = r2_score(y_test, test_preds) if len(y_test) > 1 else np.nan
else:
    mae, r2 = np.nan, np.nan

# 4. Final model on full history
model = RandomForestRegressor(n_estimators=300, random_state=42)
model.fit(X, y)
history = g.copy()
preds_map = {}
for yr in [2025, 2026, 2027]:
    recent = history["Yield"].values
    x_next = pd.DataFrame({
        "year_idx": history["year_idx"].iloc[-1] + 1,
        "lag1": recent[-1],
        "lag2": recent[-2] if len(recent) >= 2 else recent[-1],
        "roll3_mean": np.mean(recent[-3:]),
        "pct_change": (recent[-1] - recent[-2])/recent[-2] if len(recent) >= 2 and recent[-2] != 0 else 0
    })
    next_pred = model.predict(x_next)[0]
    preds_map[yr] = next_pred
    history = pd.concat([history, pd.DataFrame([{"YearNum": yr, "Yield": next_pred, "year_idx":
x_next["year_idx"].iloc[0]}]), ignore_index=True)
results.append({
    "Crop Name": crop, "Season": season, "Last_Available_Year": 2025,
    "Last_Available_Yield (2025 Pred)": round(preds_map[2025], 1),
    "Predicted_Yield_2026": round(preds_map[2026], 1),
    "Predicted_Yield_2027": round(preds_map[2027], 1),
    "Method": "Random Forest (recursive forecast)",

```

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"Test_MAE": round(mae, 1) if not np.isnan(mae) else np.nan,

"Test_R2": round(r2, 3) if not np.isnan(r2) else np.nan

})

results_df = pd.DataFrame(results).sort_values(["Crop Name", "Season"])

results_df.to_csv(OUTPUT_FILE, index=False)

display(results_df)

print(f"Trained and saved to {OUTPUT_FILE}")

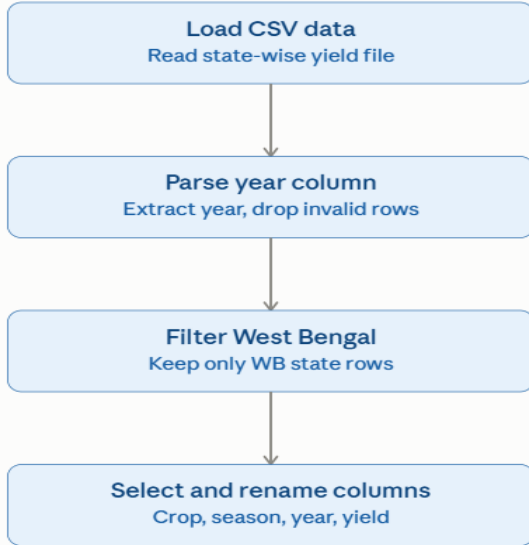


Fig-6: Block Diagram of setup and Data loading

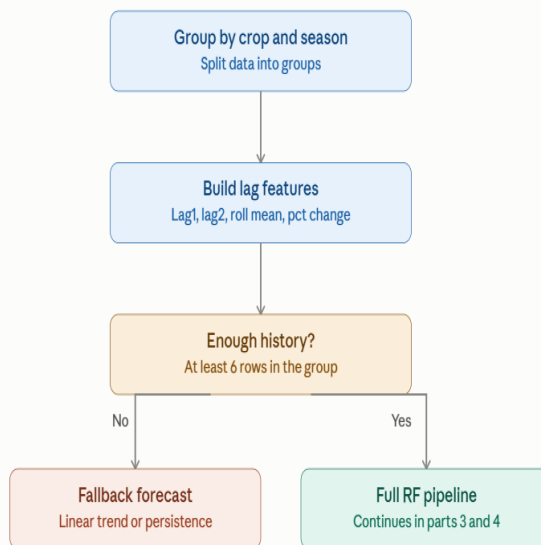


Fig-7: Block Diagram of Train+ Forecast per group

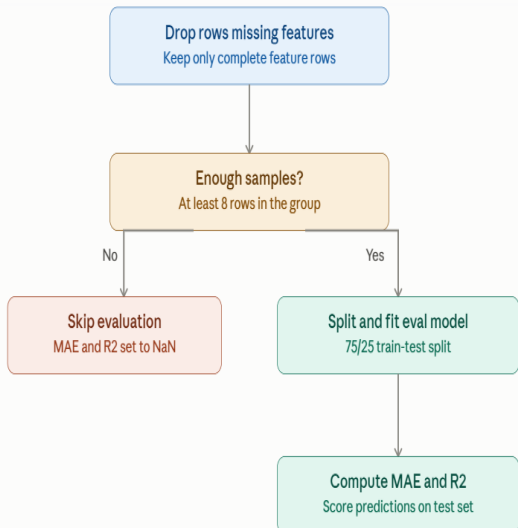


Fig-8: Block Diagram of Split for metrics

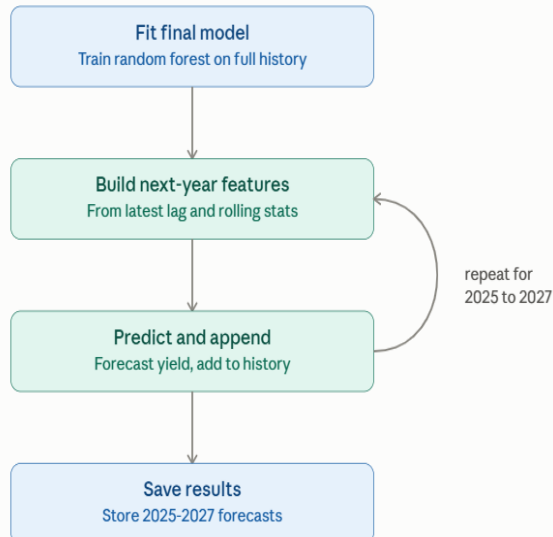


Fig-9: Block Diagram Final model on full history

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7.2 Output of predicted yield of 2026-2027:-

	Crop Name	Season	Last_Available_Year	Last_Available_Yield (2025 Pred)	Predicted_Yield_2026	Predicted_Yield_2027	Method	Test_MAE	Test_R2
0	Bajra	Total	2025	414.2	417.3	416.8	Random Forest (recursive forecast)	16.9	0.408
1	Foodgrains	Kharif	2025	2762.8	2803.1	2803.7	Random Forest (recursive forecast)	64.8	0.568
2	Foodgrains	Rabi	2025	3503.1	3466.6	3392.2	Random Forest (recursive forecast)	115.8	-0.030
3	Foodgrains	Total	2025	3078.3	3086.7	3090.2	Random Forest (recursive forecast)	60.9	0.678
4	Gram	Total	2025	1240.6	1241.5	1241.5	Random Forest (recursive forecast)	60.3	0.132
5	Groundnut	Kharif	2025	1537.5	1537.5	1537.5	Random Forest (recursive forecast)	41.7	0.122
6	Groundnut	Rabi	2025	1927.6	1980.0	2278.6	Random Forest (recursive forecast)	165.2	0.585
7	Groundnut	Summer	2025	3280.0	3280.0	3280.0	Persistence Fallback	NaN	NaN
8	Groundnut	Total	2025	2888.1	2834.5	2844.3	Random Forest (recursive forecast)	197.1	0.483
9	Linseed	Total	2025	469.9	468.0	469.6	Random Forest (recursive forecast)	36.7	0.279
10	Maize	Kharif	2025	4233.3	4150.9	4151.7	Random Forest (recursive forecast)	162.0	0.253
11	Maize	Rabi	2025	7099.0	7071.6	7037.1	Random Forest (recursive forecast)	356.5	-0.039
12	Maize	Summer	2025	7270.0	7270.0	7270.0	Persistence Fallback	NaN	NaN
13	Maize	Total	2025	6750.8	6698.9	6689.5	Random Forest (recursive forecast)	302.8	0.240
14	Oilseeds	Kharif	2025	1365.9	1365.7	1365.6	Random Forest (recursive forecast)	31.9	0.852
15	Oilseeds	Rabi	2025	1263.4	1270.1	1275.6	Random Forest (recursive forecast)	87.6	0.287
16	Oilseeds	Total	2025	1326.1	1301.6	1300.4	Random Forest (recursive forecast)	87.4	0.169
17	Pulses	Total	2025	945.5	949.7	967.0	Random Forest (recursive forecast)	60.6	0.529
18	Rapeseed & Mustard	Total	2025	1242.3	1242.2	1243.9	Random Forest (recursive forecast)	88.8	-0.088

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	Crop Name	Season	Last_Available_Year	Last_Available_Yield (2025 Pred)	Predicted_Yield_2026	Predicted_Yield_2027	Method	Test_MAE	Test_R2
19	Rice	Kharif	2025	2783.9	2837.0	2826.8	Random Forest (recursive forecast)	61.2	0.619
20	Rice	Rabi	2025	3518.1	3571.2	3565.1	Random Forest (recursive forecast)	106.1	0.048
21	Rice	Summer	2025	3617.0	3617.0	3617.0	Persistence Fallback	NaN	NaN
22	Rice	Total	2025	2974.8	3012.3	3024.9	Random Forest (recursive forecast)	64.0	0.724
23	Safflower	Total	2025	390.1	391.5	390.1	Random Forest (recursive forecast)	129.4	0.420
24	Soybean	Total	2025	772.5	782.9	778.9	Random Forest (recursive forecast)	161.8	-2.895
25	Sunflower	Kharif	2025	1667.0	1667.0	1667.0	Persistence Fallback	NaN	NaN
26	Sunflower	Rabi	2025	1277.1	1275.7	1275.3	Random Forest (recursive forecast)	69.5	0.172
27	Sunflower	Summer	2025	1150.0	1150.0	1150.0	Persistence Fallback	NaN	NaN
28	Sunflower	Total	2025	1180.7	1218.8	1222.5	Random Forest (recursive forecast)	98.2	0.097
29	Tur	Total	2025	1688.0	1651.4	1667.2	Random Forest (recursive forecast)	265.7	0.153
30	Wheat	Total	2025	3108.1	3107.7	3108.1	Random Forest (recursive forecast)	96.1	-0.161

7.3 Code and Visual Representation of Predicted Yield for 2026-2027:-

```

import matplotlib.pyplot as plt
import seaborn as sns
plot_df = results_df[['Crop Name', 'Season', 'Predicted_Yield_2026', 'Predicted_Yield_2027']]
plot_df_melted = plot_df.melt(
    id_vars=['Crop Name', 'Season'],
    value_vars=['Predicted_Yield_2026', 'Predicted_Yield_2027'],
    var_name='Predicted Year',
    value_name='Predicted Yield'
)
plt.figure(figsize=(15, 10))
sns.barplot(x='Predicted Year', y='Crop Name', hue='Predicted Year', data=plot_df_melted, palette='viridis')
plt.title('Predicted Yields for 2026 and 2027 by Crop and Season')
plt.xlabel('Predicted Year (Kg/Ha)')
plt.ylabel('Crop Name')
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.legend(title='Year')
plt.tight_layout()
plt.show()

```

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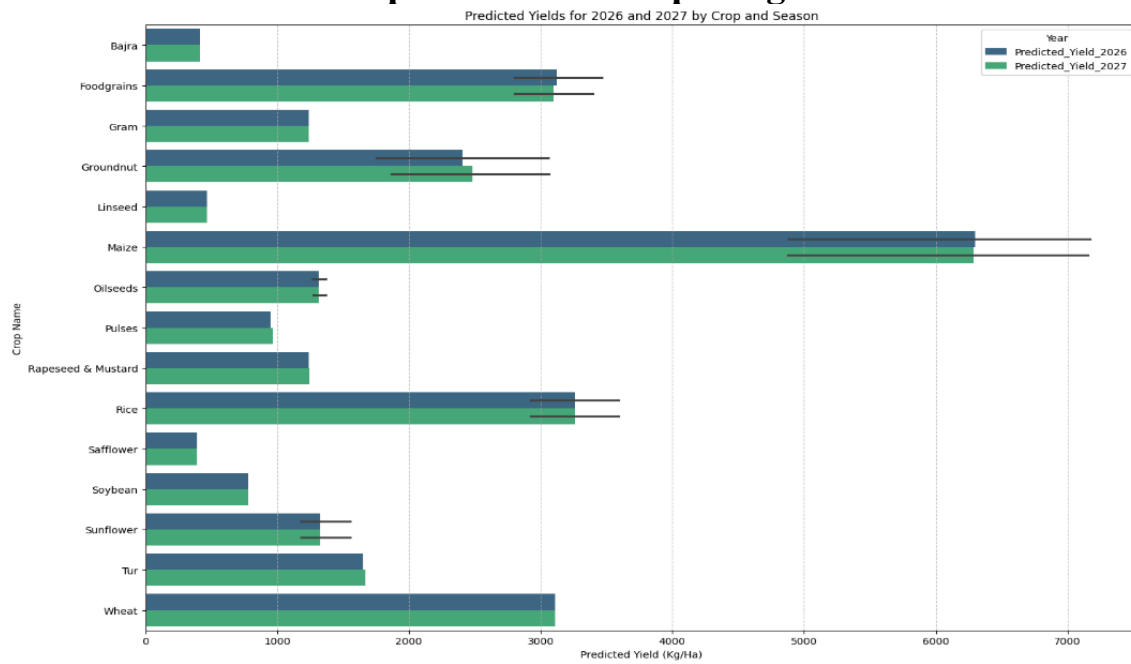


Fig 10:- Visual Representation of the Predicted Yield

7.4 General Prediction Results:-

Different tendencies can be noticed for the predictions of different crop-season combinations.

Thus, the prediction of Foodgrains – Kharif is made at the levels of 2803.1 kg/Ha in 2026 and 2803.7 kg/Ha in 2027, meaning a rather constant forecast.

As for Rice – Total, its prediction makes 3012.3 kg/Ha in 2026 and 3024.9 kg/Ha in 2027, which means an increase. The prediction of Groundnut – Rabi demonstrates a more pronounced tendency of growth from 1980.0 kg/Ha in 2026 up to 2278.6 Kg/Ha.

7.5 Performance Discussion of the Model:-

The model performed differently across the different crop-seasons. For instance, Oilseeds-Kharif provided the best recorded R^2 value at 0.852, and Rice-Total performed at 0.724. These results suggest that the designed historical yield features are quite informative for these two groups.

On the other hand, Soybean-Total recorded the worst performing R^2 at -2.895, meaning poor performance when evaluated in the subset evaluation.

The co-existence of good results and bad results suggests that the performance of historical yields-based models is highly dependent on the crop-season time series.

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8. LIMITATIONS AND FUTURE SCOPE:-

8.1 Limitations:-

There are several weaknesses present in the current study.

1. The developed model is predominantly based on historical yields and temporal features.
2. No weather-related variables are included (e.g. rainfall and temperature).
3. Soil characteristics are missing.
4. Fertilization and irrigation are not covered.
5. Not all of Crop and Season groupings have sufficient historical data.
6. Recursive forecast may carry forecast error from one year to another.
7. Prediction accuracy greatly differs depending on crop-season grouping.
8. In some groupings persistence needs to be used instead of Random Forest forecast.
9. Time-series-based validation is not employed; the evaluation is done through random split.

8.2 Future Scope:-

There are various possibilities for the expansion of the system.

1. Datasets on rainfall and temperature can be included.
2. Information about soil can be considered as another input parameter.
3. Fertilizer application and irrigation details can be included.
4. Temporal or walk forward validation approach can be utilized for future predictions.
5. Random Forest can be compared to XGBoost, Gradient Boosting, Extra Trees and Artificial Neural Network.
6. Hyperparameter tuning can be implemented.
7. Feature importance analysis can be performed to recognize impactful historical features.
8. Prediction intervals can be calculated to visualize forecast uncertainty.
9. System can be developed up until the year 2027.
10. A web-based forecasting interface can be developed.

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9. CONCLUSION:-

A Random Forest Regression based framework has been used in order to predict the crop yields for the state of West Bengal for the years 2026 and 2027.

The past records have been pre-processed in order to select the data related to West Bengal. Then the past agricultural data were arranged in Crop name and Season, and the time-based features were prepared like the year index, lags, rolling means, and the percent change.

Random Forest Regression Models have been trained using the groups which had enough data in terms of Crop and Season and have been checked using Mean absolute error and R-squared scores. It has been seen that depending upon the groups, the score varied to a significant extent. For example, Oilseeds-Kharif is known to have got the highest R-square score of 0.852, while other crops got negative or poor R-squared scores.

The prediction process has then been used for future prediction of crops for the years 2026 and 2027.

However, the existing model excludes factors like weather, soil conditions, fertilizers, water supply, and many others from agriculture.

In general, this paper illustrates how Random Forest regression can be used as a methodological tool to estimate the crop yield on the basis of historical patterns with clear potential for further improvements of the model's performance.

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